Muhammad Arslan Aslam

Sex: Male | **Phone:** (+34) 613233257 (Mobile) | **Email:** muhammad-arslan.aslam@ufz.de |

Website: https://www.researchgate.net/profile/Muhammad-Arslan-Aslam/research |

Address: Friedrich-Ebert-Strabe, 82, 04109, Leipzig, Germany (Home)

WORK EXPERIENCE

01/01/2024 – CURRENT Leipzig, **Germany**

RESEARCHER HELMHOLTZ CENTRE FOR ENVIRONMENTAL RESEARCH

DiPEM Project (Digital Molecular Endocrine Profiling)

1. Conducting **zebrafish embryo (ZFE) toxicity assays** to assess the effects of **endocrine-disrupting chemicals (EDCs)**.
2. Contributing to the development of mechanistically anchored **digital biomarkers** by linking phenotypic effects with **AOPs** to expand and refine the Fish Early Life Stage Test (FELS).
3. Supporting the replacement of adult-based FSTRA with embryo-based **NAMs**, aligning with 3Rs and OECD guidance on endocrine disruptor testing.

EU PrecisionTox Project (Mechanistic Toxicology Using New Approach Methods (NAMs))

1. Contributed to **high-throughput phenotypic** screening of 200 chemicals using zebrafish embryos (ZFE) to assess adverse outcomes relevant to human health.
2. Employed automated image-based analyses to evaluate **morphological, cardiovascular, and behavioral endpoints** (e.g., photomotor response, spontaneous movement), including those potentially linked to neuro- and endocrine activity.
3. Assessed specificity of chemical effects by calculating sensitivity ratios (**predicted baseline EC50** / measured EC50)
4. Used effect specificity patterns to support **mode-of-action (MoA)** inference and improve predictive toxicology for structurally uncharacterized compounds especially endocrines.

EU PARC Project (Partnership for the Assessment of Risks from Chemicals)

1. Assessed endocrine-disruptive potential of **BPA** and four structural analogues (BPE, BPAP, BPZ, BPG) by morphological, cardiovascular, and behavioral effects of zebrafish embryos (ZFE).
2. Developed novel phenotypic endpoints and trained **Python-based models** for automated annotation of morphology.
3. Applied PrecisionTox specificity framework (baseline EC50 / observed EC50 >10) to identify specific-acting compounds; BPE showed the highest specificity and overall toxicity.

01/01/2023 – 30/06/2023 Bilbao, **Spain**

RESEARCH INTERNEE UPV/EHU PIE MARINE STATION

1. Prepared Water Accommodated Fractions using heavy and light oils for **Sea urchin embryo toxicity**
2. Significant reductions in fertilization success and larval development observed
3. Identifying and quantifying chemicals using **LC-HRMS**

EDUCATION AND TRAINING

01/09/2022 – 01/08/2024 Europe

ERASMUS MUNDUS MSC IN MARINE ENVIRONMENT AND RESOURCES

Co-funded by the Erasmus+ Programme of the European Union

1. University of Bordeaux, France **(1st Semester)**
2. University of the Basque Country, Spain - UPV/EHU **(2nd Semester)**
3. University of Liege, Belgium **(3rd Semester)**
4. Helmholtz Centre for Environmental Research**,** Germany **(Master Thesis)**

Thesis: COMPARATIVE TOXICITY ASSESSMENT OF BISPHENOLS ADDITIVES USING A HIGH CONTENT ZEBRAFISH ASSAY

1. This study evaluated the developmental toxicity of BPA and its substitutes (BPE, BPAP, BPZ, BPG) in zebrafish embryos using multi-endpoint, high-content screening approach.

2. The results demonstrated that BPE exhibited the highest toxicity, affecting multiple endpoints, including neurological and cardiovascular development.

3. This research provides important insights for the regulatory assessment of bisphenol substitutes and their potential health risks.

BACHELOR IN ENVIRONMENTAL SCIENCES - 3.84/4 (BRONZE MEDALIST) Government College University Faisalabad

Thesis: Awareness, Disposal Behaviors, Willingness, and Inflow of Electronic Waste in Faisalabad, Pakistan

1. Designed and conducted a mixed-methods study (surveys, field visits, and questionnaires)
2. 75.2% of respondents were unaware of e-waste; 82% lacked knowledge of its hazards; 45% relied on informal recycling; 91% showed willingness to participate in formal management schemes.
3. Contributed to sustainable waste management discourse by providing data-driven evidence for targeted interventions and policy development in Pakistan.

SPECIALIZATION IN CONCEPTS OF TOXICOLOGY AND CHEMICALS Jhons Hopkins University

Toxicology 21: Scientific Applications
- Concepts of Pathways of Toxicity (PoT)

Chemicals and Health
- Risk Assessment of Chemicals
- Regulatory Framework and

Evidence-based Toxicology
- Systematic Review and Meta-Analysis
- Biases in Conducting Toxicity Assays

DIGITAL SKILLS

Experience in python, and R | Reference Management in Zotero, Mendeley | Software Tools: KNIME and other common applications (e.g. Microsoft Office, LaTeX)

	R	Python	KNIME
Data Processing	●●●●●	●●●○○	●●●●●
Statistical Analysis	●●●●○	●●●○○	●●●●○
Visualization	●●●●●	●●●●○	●●●●●
Machine Learning	●●●○○	●●○○○	●●○○○

PUBLICATIONS

Research Article(s)

Alghanem, S. M. S., Alsudays, I. M., Farid, M., Sarfraz, W., Ishaq, H. K., Farid, S., Zubair, M., Khalid, N., **Aslam, M. A.**, Abbas, M., & Abeed, A. H. A. (2024). **Evaluation of heavy metal accumulation and tolerance in oxalic acid-treated Phragmites australis wetlands for textile effluent remediation**. International Journal of Phytoremediation, 1–16. https://doi.org/10.1080/15226514.2024.2372849

Aslam, M. A., Ahmad, M. M., Farid, S., & Hashmi, S. I. (2025). **Awareness, disposal behaviours, willingness, and inflow of electronic waste in Faisalabad, Pakistan.** International Journal of Environment and Waste Management, 36(4), 472–490. https://doi.org/10.1504/ijewm.2025.144994

Book Chapter(s)

Salimkumar A.J., Cleetus M.C.K., Essel D.A., **Aslam M.A.,** Galib A.H., Prabhakaran M.P., and Rejish K.V.J (2024) **Plastic Peril: Unraveling Microplastics Threats to Health and Ecosystems**. In: Singh N.K., Jayaswal K., Yadav M., Maharjan N., editors. Fate of Microplastics in Wastewater Treatment Plants Occurrence, Identification, Potential Factors, and Future Perspectives. 1st ed. Tylor & Francis: CRC Press (22 November, 2024). https://www.routledge.com/Fate-of-Microplastics-in-Wastewater-Treatment-Plants-Occurrence-Identification-Potential-Factors-and-Future-Perspectives/Singh-Jayaswal-Yadav-Maharjan/p/book/9781032538778

Farid, M., Ahsan, A., Asam, Z. U. Z., Abbas, M., Fatima, A., Salman, M., Tariq, S., Iqbal, A., Akhtar, W., Ali, S., & **Aslam, M**. (2023). **Principles and Applications of Environmental Biotechnology for Sustainable Future.** In Springer eBooks (pp. 751–775). https://doi.org/10.1007/978-3-031-37424-1_34

Farid, M., Farid, S., **Aslam, M.**, Azhar, K., Tariq, K., Ali, Z., Ijaz, Z., Sarfraz, Z., Khalid, M., & Ahmad, M. (2023). **Biotechnological techniques for sustainable waste management.** In Springer eBooks (pp. 689–712). https:// doi.org/10.1007/978-3-031-37428-9_30

HONOURS AND AWARDS

Her Majesty Queens Thailand Scholarship – Asian institute of Technology, Thailand

Award of Laptop on Securing 2nd position in university – Ministry of Education, Pakistan

European Erasmus Mundus Scholarship –

CONFERENCES & SEMINARS

Presenter, PrecisionTox Annual Consortium Meeting, Brussels, Belgium, June 13–14, 2024.

Poster Presenter, PARC Consortium Meeting, Hall in Tirol, Austria, May 13–16, 2024.

Project Presenter, Asian Institute of Technology Sustainability Hackathon 2021

RECOMMENDATIONS

Dr. Stefan Scholz Project & Thesis Supervisor
Department of Ecotoxicology - UFZ, Leipzig, Germany
Email stefan.scholz@ufz.de **Phone** (+49) 34160251080

Dr. Wibke Busch Project Supervisor
Department of Ecotoxicology – UFZ, Leipzig, Germany
Email wibke.busch@ufz.de **Phone** (+49) 34160251581

2020-2026

With the support of the Erasmus+ Programme of the European Union

université de BORDEAUX · UNIVERSITY OF Southampton · Universidad del País Vasco / Euskal Herriko Unibertsitatea · LIÈGE université · UAc UNIVERSIDADE DOS AÇORES

THE MARINE ENVIRONMENT & RESOURCES (MER) CONSORTIUM

UNIV SOUTHAMPTON, UNIV BORDEAUX, UNIV LIEGE, UNIV PAIS VASCO/EUSKAL HERRIKO UNIB, UNIV AÇORES

CERTIFIES THAT

Muhammad Arslan Aslam

HAS SURPASSED THE 120 ECTS CORRESPONDING TO THE

Erasmus Mundus Joint Master Degree in Marine Environment MER2030 EMJMD

PLENTZIA, BASQUE COUNTRY (Spain).

SEPTEMBER 11, 2024

Note: Provisional notification only – awaiting confirmation from participating partners

Prof. Cecilia D'Angelo	Prof. Jörg Schäfer	Prof. Ionan Marigómez	Prof. Manu Soto	Prof. Sylvie Gobert	Prof. Patrica Garcia
Univ Southampton	Univ Bordeaux	UPV/EHU; MER MSc Coordinator	UPV/EHU	Univ. Liège	Univ. Açores

Erasmus Mundus Joint Master Degree in Marine Environment
www.mer2030.eu

Co-funded by the Erasmus+ Programme of the European Union

université de BORDEAUX · University of Southampton · LIÈGE université · Universidad del País Vasco / Euskal Herriko Unibertsitatea

European MSc in Marine Environment

Erasmus Mundus Joint Master Degree 619798/2020 - MER2030 EMJMD

Cohort	2022-2024	Acad Years	2022-23 & 2023-24
		Graduation date	09/09/2024
Average grade	B		

Name: ASLAM, MUHAMMAD ARSLAN

Passport: SSSSSS

		MODULE	ECTS	UBx	EHU	EHU-4	SOTON	LU	GRADE
YEAR 1 SEMESTER 1 PATHWAY A	Universtité de Bordeaux	Methodological basis for Environmental Sciences (Analysis and Modelling)	6	7,60	5,00	1,00	50	10	E
		Biological Oceanography	6	12,15	6,74	1,00	64	14	C
		Dynamic Oceanography	6	11,05	6,03	1,00	60	13	C
		Chemical Oceanography	6	10,85	5,85	1,00	58	12	D
		Seafloor Geology	6	11,73	6,46	1,00	62	13	C

AVERAGE MARK	EHU	EHU-4	UBx	SOTON	ULiège
	6,02	1,00	10,676	59	12

				SOTON	EHU	EHU-4	UBx	LU	GRADE
YEAR 1 SEMESTER 1 PATHWAY B	Univ Southampton	Introduction to Biological Oceanography	3,75						
		Introduction to Physical Oceanography	3,75						
		Introduction to Chemical Oceanography	3,75						
		Introduction to Marine Geology	3,75						
			7,5						
			7,5						

AVERAGE MARK	EHU	EHU-4	UBx	SOTON	ULiège
	-	-	-	-	-

				EHU	UBX	EHU-4	SOTON	LU	GRADE
YEAR 1 SEMESTER 2	Universidad del Pais Vasco / Euskal Herriko Unibertsitatea	Research in Marine Environment and Resources (RiMER)	6	9,00	15,91	3,00	72	17	A
		Socio-economic aspects of climate change	4	8,00	14,11	2,00	67	16	B
		Ocean Global Change Biology	4	7,00	12,56	2,00	65	15	B
		Degradation and Rehabilitation of Estuarine Ecosystems	4	9,00	15,91	3,00	72	17	A
		Ecosystem-based Fisheries Management	4	8,00	14,11	2,00	67	16	B
		Environmental Monitoring and Risk Assessment of Aquatic Systems	4	7,70	13,64	2,00	66	15	B
		Satellite Oceanography and Meteorology	4	7,70	13,64	2,00	66	15	B

AVERAGE MARK	EHU	EHU-4	UBx	SOTON	ULiège
	8,12	2,33	14,378	68	16

				SOTON	EHU	EHU-4	UBx	LU	GRADE
YEAR 2 SEMESTER 3 PATHWAY A1	Univ Southampton		7,5						
			7,5						
			7,5						
			7,5						

AVERAGE MARK	EHU	EHU-4	UBx	SOTON	ULiège
	-	-	-	-	-

				LU	EHU	UBx	EHU-4	SOTON	GRADE
YEAR 2 SEMESTER 3 PATHWAYS A2 & B	Uiversité de Liège	Marine Ecology	6	13	6,00	11,00	1,00	59	C
		Ecotoxicology and Biodegradation of Marine Pollutants	6	15	7,00	12,56	2,00	65	B
		Remote Sensing of the Oceans	6	13	6,00	11,00	1,00	59	C
		Marine Plant Biology and Ecology	6	16	8,00	14,11	2,00	67	B
		Professional Practice	6	18	9,33	17,26	3,00	81	A

AVERAGE MARK	EHU	EHU-4	UBx	SOTON	ULiège
	7,27	1,80	13,186	66	15

				EHU	UBx	EHU-4	SOTON	LU	GRADE
YEAR 2 SEM 4	UPV/EHU	Master (MSc) Thesis *	30	8,50	14,88	2,00	68	16	B

AVERAGE MARK	EHU	EHU-4	UBx	SOTON	ULiège
	8,50	2,00	14,885	68	16

IMPORTANT: INTRODUCE MARKS (AND MODULE INDENTIFIERS) IN THE WHITE BOXES. USING "PASTE VALUES"

TOTAL ECTS (2022-2023 & 2023-2024): 120

Pathway A1: U Bordeaux (S1) + U Basque Country (S2) + U Southampton (S3) + MSc Thesis (S4)
Pathway A2: U Bordeaux (S1) + U Basque Country (S2) + U Liège (S3) + MSc Thesis (S4)
Pathway B: U Southampton (S1) + U Basque Country (S2) + U Liège (S3) + MSc Thesis (S4)

Pathway	EHU	EHU-4	UBx	SOTON	ULiège
A1	-	-	-	-	
A2	7,48	1,78	13,281		15
B	-	-		-	-

* **Title:** Comparative toxicity assessment of plastic bisphenols additives and analogs using a high content zebrafish assay
Supervisor: Stefan Scholz
Institution: Helmholtz-Zentrum für Umweltforschung (UFZ), Leipzig (Germany)

Rank*: 50

* Percentage of students with lower average mark

2024/9/11

Ionan MARIGOMEZ
MER2030 EMJMD Prog Coord MER Consortium JPB Chair

Universidad del País Vasco / Euskal Herriko Unibertsitatea · PiE Plentzia Itsas Estazioa / Estación Marina de Plentzia

MER Consortium SECRETARIAT - PIE-UPV/EHU
e-mail: mer@merconsortium.eu

رجسٹریشن نمبر
Registration Number
2016-GCUF-04135

بِسۡمِ اللّٰہِ الرَّحۡمٰنِ الرَّحِیۡمِ

رول نمبر
Roll Number
20033

Government College University Faisalabad, Pakistan

This University hereby confers upon

یہ جامعہ

Muhammad Arslan

محمد ارسلان

S/o / D/o *Muhammad Aslam*

بن / بنت محمد اسلم

of *Department of Environmental Sciences and Engineering*

از ڈیپارٹمنٹ آف انوائرنمنٹل سائنسز اینڈ انجینئرنگ

the Degree of *BS*

کو بی ایس

Environmental Science

انوائرنمنٹل سائنس

with all the rights, honours and privileges pertaining to this degree

کی سند مع تمام حقوق، امتیازات اور اعزازات جو اس سند کے ساتھ وابستہ ہیں عطا کرتی ہے

Session: 2016-2020

دورانیہ: ۲۰۱۶-۲۰۲۰ء

CGPA: 3.84 / 4.00

سی جی پی اے: ۳.۸۴\۴.۰۰

Result Declaration Date: 05-09-20

تاریخ اعلان نتیجہ: ۰۵-۰۹-۲۰

011273

M Sarwar

Chancellor چانسلر

Controller of Examinations کنٹرولر امتحانات

Government College University
Faisalabad, Pakistan

TRANSCRIPT

BS Environmental Science, SESSION 2016-2020

NAME: MUHAMMAD ARSLAN
FATHER NAME: MUHAMMAD ASLAM

REG. NO: 2016-GCUF-04135
ROLL NO: 20033

Sr. No.	Course Code	Subject Title	Credit Hours	Max. Marks	Obt. Marks	Marks Per (%)	Letter Grade	Q.P.	Remarks
1st Semester									
1	CHM-305	BASIC CHEMISTRY	3(2-1)	60	48.00	80.00	A	12.00	
2	ENG-321	FUNCTIONAL ENGLISH	3(3-0)	60	45.00	75.00	B	11.00	
3	ENV-301	INTRODUCTION TO ENVIRONMENTAL SCIENCES	3(3-0)	60	52.25	87.08	A	12.00	
4	ENV-303	BIOLOGY-I	3(2-1)	60	48.50	80.83	A	12.00	
5	MTH-321	MATHEMATICS-I	3(3-0)	60	48.00	80.00	A	12.00	
6	PST-321	PAKISTAN STUDIES	2(2-0)	40	30.00	75.00	B	7.33	
		Semester Total :	**17**	**340**	**271.75**	**79.93**		**66.33**	**GPA, 3.90**
2nd Semester									
1	ENG-322	ENGLISH COMPREHENSION AND COMPOSITION	3(3-0)	60	46.00	76.67	B	11.33	
2	ENV-302	INTRODUCTION TO EARTH SCIENCES	3(2-1)	60	52.50	87.50	A	12.00	
3	ENV-304	BIOLOGY-II	3(2-1)	60	48.00	80.00	A	12.00	
4	ENV-306	ENVIRONMENTAL ANTHROPOLOGY	3(3-0)	60	51.00	85.00	A	12.00	
5	ISL-321	ISLAMIC STUDIES	2(2-0)	40	35.00	87.50	A	8.00	
6	STT-321	INTRODUCTION TO STATISTICAL THEORY	3(3-0)	60	53.75	89.58	A	12.00	
		Semester Total :	**17**	**340**	**286.25**	**84.19**		**67.33**	**GPA, 3.96**
3rd Semester									
1	CSI-321	INTRODUCTION TO COMPUTING APPLICATIONS	3(1-2)	60	48.00	80.00	A	12.00	
2	ECO-421	ISSUES IN PAKISTAN ECONOMY	3(3-0)	60	50.00	83.33	A	12.00	
3	ENG-421	COMMUNICATION SKILLS	3(3-0)	60	42.00	70.00	B	10.00	
4	ENV-401	ENVIRONMENTAL CHEMISTRY	3(2-1)	60	43.00	71.67	B	10.33	
5	ENV-403	ENVIRONMENTAL PHYSICS	3(2-1)	60	39.00	65.00	B	9.00	
6	ENV-405	PHILOSOPHY	3(3-0)	60	48.00	80.00	A	12.00	
		Semester Total :	**18**	**360**	**270**	**75.00**		**65.33**	**GPA, 3.63**
4th Semester									
1	ENV-402	FUNDAMENTALS OF ECOLOGY	3(3-0)	60	48.00	80.00	A	12.00	
2	ENV-404	ENVIRONMENTAL MICROBIOLOGY	3(2-1)	60	49.00	81.67	A	12.00	
3	ENV-406	ENVIRONMENTAL POLLUTION	3(3-0)	60	48.75	81.25	A	12.00	
4	ENV-408	CLIMATOLOGY	3(3-0)	60	50.25	83.75	A	12.00	
5	ENV-410	WASTE MANAGEMENT	3(3-0)	60	50.00	83.33	A	12.00	
6	PSY-422	INTRODUCTION TO PSYCHOLOGY	3(3-0)	60	49.00	81.67	A	12.00	
		Semester Total :	**18**	**360**	**295**	**81.94**		**72.00**	**GPA, 4.00**
5th Semester									
1	ENV-501	APPLIED ECOLOGY	3(2-1)	60	48.80	81.33	A	12.00	
2	ENV-503	BASICS OF ENVIRONMENTAL TOXICOLOGY	3(2-1)	60	46.50	77.50	B	11.50	
3	ENV-505	ENVIRONMENTAL PROFILE OF PAKISTAN	3(3-0)	60	48.00	80.00	A	12.00	
4	ENV-507	FUNDAMENTALS OF ENVIRONMENTAL ECONOMICS	3(3-0)	60	48.00	80.00	A	12.00	
5	ENV-509	ANALYTICAL TECHNIQUES IN ENVIRONMENTAL SCIENCE	3(1-2)	60	51.00	85.00	A	12.00	
6	ENV-511	ENERGY AND ENVIRONMENT	3(3-0)	60	53.00	88.33	A	12.00	
		Semester Total :	**18**	**360**	**295.3**	**82.03**		**71.50**	**GPA, 3.97**
6th Semester									
1	ENV-502	FUNDAMENTALS OF ENVIRONMENTAL BIOTECHNOLOGY	3(2-1)	60	45.00	75.00	B	11.00	
2	ENV-504	GIS AND REMOTE SENSING	3(2-1)	60	45.25	75.42	B	11.10	

PREPARED BY :

CHECKED BY :

DY. CONTROLLER :

RESULT DECLARATION DATE : 05-SEP-20

DATE OF ISSUE : SEPTEMBER 6, 2020, 3:08 PM

SERIAL N0 : 191694, 191695

CONTROLLER OF EXAMINATIONS
GC UNIVERSITY, FAISALABAD

Government College University
Faisalabad, Pakistan

TRANSCRIPT

BS Environmental Science, SESSION 2016-2020

NAME: MUHAMMAD ARSLAN **REG. NO:** 2016-GCUF-04135

FATHER NAME: MUHAMMAD ASLAM **ROLL NO:** 20033

Sr. No.	Course Code	Subject Title	Credit Hours	Max. Marks	Obt. Marks	Marks Per (%)	Letter Grade	Q.P.	Remarks
3	ENV-506	BASICS OF ENVIRONMENTAL MANAGEMENT SYSTEMS	3(3-0)	60	41.00	68.33	B	9.67	
4	ENV-508	BIODIVERSITY AND CONSERVATION	3(3-0)	60	36.00	60.00	C	8.00	
5	ENV-510	ENVIRONMENTAL MONITORING	3(2-1)	60	42.00	70.00	B	10.00	
6	ENV-512	WATER RESOURCE MANAGEMENT	3(3-0)	60	48.00	80.00	A	12.00	
		Semester Total :	**18**	**360**	**257.25**	**71.46**		**61.77**	**GPA, 3.43**
7th Semester									
1	ENV-601	INTRODUCTION TO CLIMATE CHANGE	3(3-0)	60	48.00	80.00	A	12.00	
2	ENV-603	BASICS OF ENVIRONMENTAL IMPACT ASSESSMENT	3(3-0)	60	48.50	80.83	A	12.00	
3	ENV-605	NATURAL RESOURCE MANAGEMENT	3(3-0)	60	52.00	86.67	A	12.00	
4	ENV-607	RESEARCH METHODS IN ENVIRONMENTAL SCIENCE	3(3-0)	60	46.00	76.67	B	11.33	
5	ENV-609	URBAN PLANNING AND MANAGEMENT	3(3-0)	60	50.00	83.33	A	12.00	
6	ENV-611	SOIL AND ENVIRONMENT	3(3-0)	60	40.50	67.50	B	9.50	
		Semester Total :	**18**	**360**	**285**	**79.17**		**68.83**	**GPA, 3.82**
8th Semester									
1	ENV- 606	POLLUTION CONTROL TECHNOLOGIES	3(2-1)	60	53.20	88.67	A	12.00	
2	ENV-602	ENVIRONMENTAL LAWS AND POLICIES	3(3-0)	60	48.00	80.00	A	12.00	
3	ENV-604	HEALTH AND ENVIRONMENT	3(3-0)	60	48.20	80.33	A	12.00	
4	ENV-610	ATMOSPHERIC SCIENCE	3(3-0)	60	48.00	80.00	A	12.00	
5	ENV-612	DISASTER MANAGEMENT	3(3-0)	60	48.00	80.00	A	12.00	
		Semester Total :	**15**	**300**	**245.4**	**81.80**		**60.00**	**GPA, 4.00**
		Grand Total :	**139**	**2780**	**2205.95**	**79.35**		**533.09**	
Cumulative Grade Point Average (CGPA) required 2.5, Earned 3.84									

(ERRORS AND OMISSIONS EXCEPTED)

Certified that the candidate has successfully completed his/her degree requirements.
This Transcript is valid when signed by the Controller of Examinations alongwith official seal.

PREPARED BY :

CHECKED BY :

DY. CONTROLLER :

RESULT DECLARATION DATE : 05-SEP-20

DATE OF ISSUE : SEPTEMBER 6, 2020, 3:08 PM

SERIAL N0 : 191694, 191695

CONTROLLER OF EXAMINATIONS
GC UNIVERSITY, FAISALABAD

JOHNS HOPKINS UNIVERSITY

COURSE CERTIFICATE

08/15/2020

Arslan Aslam

has successfully completed

Evidence-based Toxicology

an online non-credit course authorized by Johns Hopkins University and offered through Coursera

EDUCATION FOR EVERYONE · coursera · COURSE CERTIFICATE

Thomas Hartung
Professor
Environmental Health and Engineering

Lena Smirnova
Research Associate
Center for Alternatives to Animal Testing

Verify at coursera.org/verify/YNVFD28AMABP

Coursera has confirmed the identity of this individual and their participation in the course.

Some online courses may draw on material from courses taught on campus but are not equivalent to on-campus courses. This certificate does not affirm that this learner was enrolled as a student at Johns Hopkins University in any way. It does not confer a JHU grade, course credit or degree; establish any relationship between this learner and JHU or other JHU affiliate; enroll or register this learner at JHU or other JHU affiliate or in any course offered by JHU; or entitle this learner to access or use the resources of JHU or other JHU affiliates beyond the online courses provided by Coursera.

JOHNS HOPKINS UNIVERSITY

COURSE CERTIFICATE

Oct 5, 2020

Arslan Aslam

has successfully completed

Toxicology 21: Scientific Applications

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